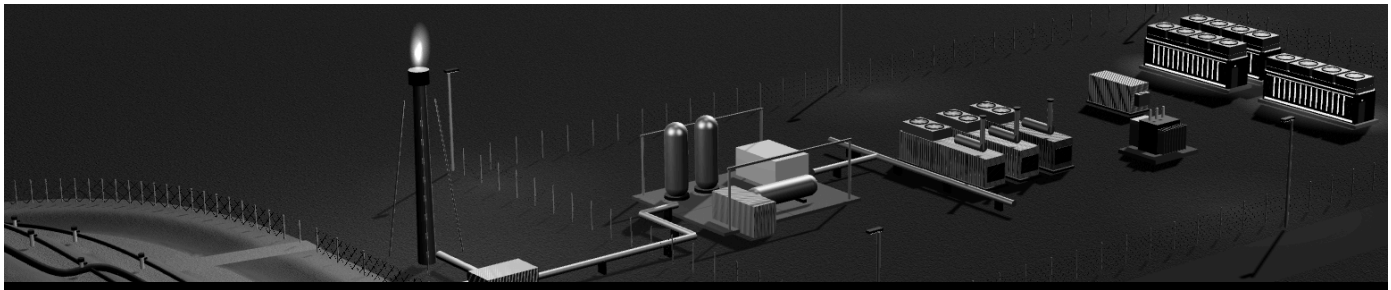
# T.E. Maxson Wastewater Treatment Facility, Memphis, Tennessee: a first sizing

What the gas the site now flares could power, how much land it would take, and what it could earn each year.

| 2.6 MW | 3 | 8 ac | 235k |
|---|---|---|---|
| POWER FROM THE GAS | COMPUTE MODULES | LAND NEEDED | MMBTU BOUGHT / YR |

| GAS SALE, PER YEAR | LAND LEASE, PER YEAR | TOTAL TO THE OWNER, PER YEAR |
|---|---|---|
| up to $176k | $8k to $16k | $8k to $193k |



YOUR SITE IN OUR LIVE 3D MODEL — ACCRETIONCOMPUTE.COM/SITE-MODEL?SRC=LFG&FLOW=950&CH4=49

## THE MATH

| | |
|---|---|
| Lagoon biogas flared (2018 onward) | 950 scfm |
| Methane (from about 500 Btu/cf) | 49% |
| Energy in the gas | 28.3 MMBtu per hour |
| Power at a heat rate of 11,000 Btu/kWh | about 2.6 MW |
| Compute (power less cooling and losses, /1.2) | about 2.1 MW |
| Compute modules (about 1 MW each) | 3 |
| Gas engines (running plus one spare) | 3 at 1.6 MW |
| Land: 5 acres plus 1.2 acres per MW | about 8 acres |
| Gas bought per year (95% run time) | 235k MMBtu |
| Gas payment, up to $0.75/MMBtu | up to $176k |
| Land lease at $1,000 to $2,000 per acre per year | $8k to $16k |

Biogas: methane at about 1,012 Btu per cubic foot. Prices in US dollars; the price band is the one shown on our Site model page.

## WHAT IT WOULD LOOK LIKE

- Gas engines on a small parcel near the gas, a battery to smooth the load, and factory-built compute modules leased to an AI company.
- No grid connection and no export: all of the power is used on site.
- We build and run the site.
- Multi-year gas and land agreements, matched to the tenant's lease, at a fixed price.

### WORTH CHECKING

- Whether the existing biogas offtake contract is still in force
- High H2S: gas cleanup sized for it
- City data center rules (Ordinance 5986, October 2026)

**Source:** Shelby County Health Department, Title V renewal statement of basis (2024): about 500 MMcf a year of lagoon biogas since 2018 (about 950 scfm) at about 500 Btu/cf, all to five flares since the pipeline offtake stopped in 2017. This is a first read from public data, not an offer: no cost and no obligation. The real price and terms are set in the gas and land agreements after a site review.